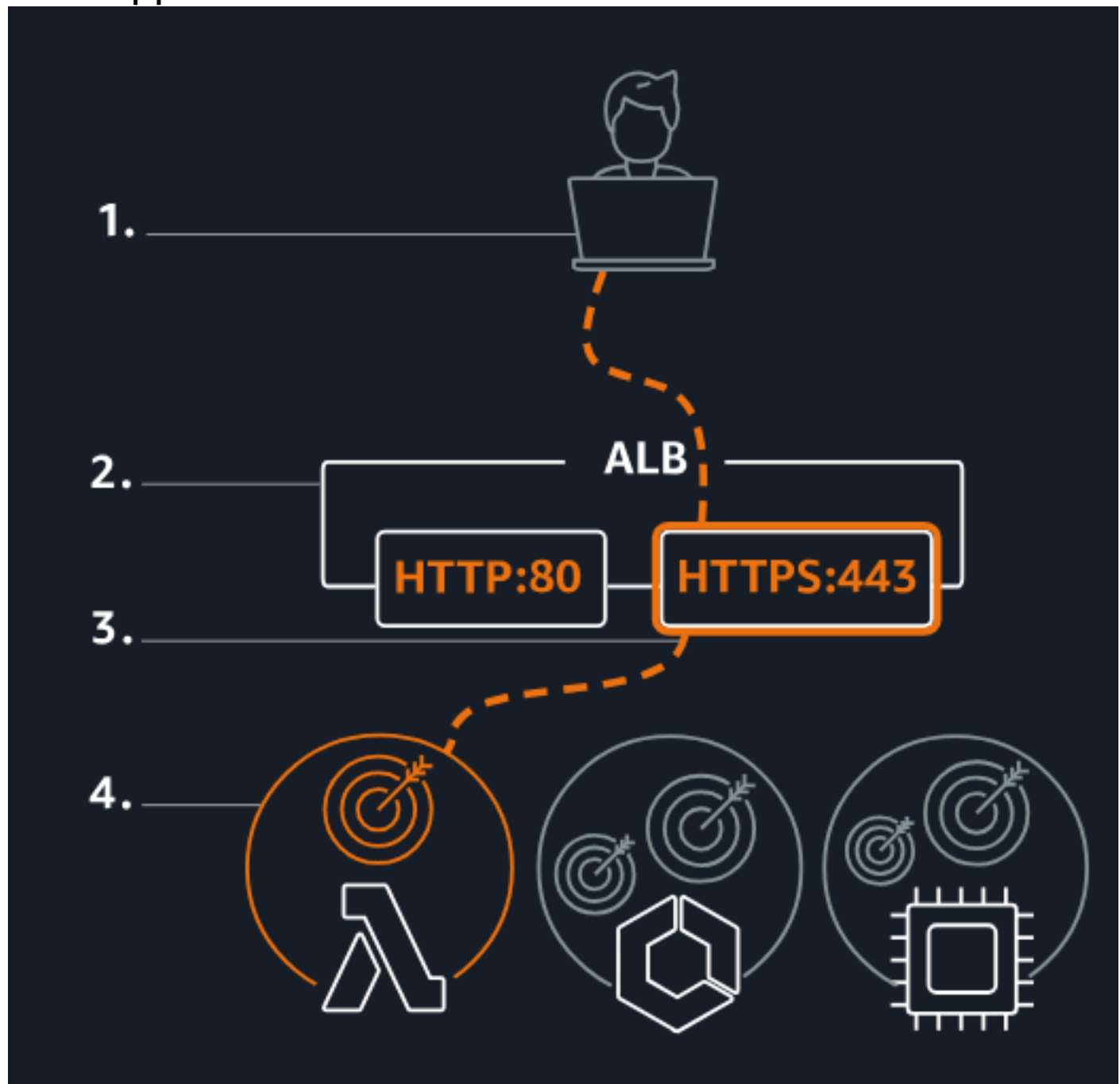


Phase 4: ALB Creation with Forced HTTPS

How Application Load Balancers work



1. Clients make requests to your application.
2. The listeners in your load balancer receive requests matching the protocol and port that you configure.
3. The receiving listener evaluates the incoming request against the rules you specify, and if applicable, routes the request to the appropriate target group. You can use an HTTPS listener to offload the work of TLS encryption and decryption to your load balancer.
4. **Healthy targets** in one or more target groups receive traffic based on the load balancing algorithm, and the routing rules you specify in the listener.

(A): Basic Config:

EC2 > Load balancers > Create Application Load Balancer

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

alb-boutiquemode

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | Info
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type | Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.

Network mapping

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC | Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0fd3d704c387ff27d
172.31.0.0/16 (default) [Create VPC](#)

IP pools | Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ **Use IPAM pool for public IPv4 addresses**
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets | Info
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az4)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-080319664ca771bd3
IPv4 subnet CIDR: 172.31.16.0/20

☒ **us-east-1b (use1-az6)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0e12576e7340a7115
IPv4 subnet CIDR: 172.31.32.0/20

☒ **us-east-1c (use1-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0b96f5c5de9d42c4d
IPv4 subnet CIDR: 172.31.0.0/20

☐ **us-east-1d (use1-az2)**

select at least two Availability Zone + Their public subnets

(B): Specific Security Group Creation

→ Check Sub Node 1

(A'): Select the newly created Security Group

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Q |

- ☐ launch-wizard-1
sg-01657dc434fb21b9e VPC: vpc-0fd3d704c387ff27d
- ☐ alb-boutiquemode
sg-02bcd3894a85b86a8 VPC: vpc-0fd3d704c387ff27d
- ☐ default
sg-09fc5679a79c78606 VPC: vpc-0fd3d704c387ff27d
- ☒ alb-sg
sg-0b856a02f7779d7e4 VPC: vpc-0fd3d704c387ff27d
- ☐ boutiquemode-web-sg
sg-0dbe92e65d5fff028 VPC: vpc-0fd3d704c387ff27d

Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTPS:443

Remove

Protocol

HTTPS

Port

443

1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Pre-routing action [Info](#)

☒ No pre-routing action

☐ Authenticate user

User auth with OIDC or Amazon Cognito

☐ Validate token

Validate client-presented JSON Web Token (JWT)

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

boutiquemode-tg

Target type: Instance, IPv4 | Target stickiness: Off

HTTP



Weight

1

0-999

Percent

100%

(C): Certificate Creation

→ Check Sub Node 2

(A"):

EC2 > Load balancers > Create Application Load Balancer

Secure listener settings

Info

These settings will apply to all of your secure listeners. Once created, you can manage these settings per listener.

Security policy

Info

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration called a security policy to manage SSL connections with clients. [Compare security policies](#)

Security category

All security policies

Policy name

New

ELBSecurityPolicy-TLS13-1-2-Res-PQ-2025-09 (Recommended)

Default SSL/TLS server certificate

The certificate used if a client connects without SNI protocol, or if there are no matching certificates. You can source this certificate from AWS Certificate Manager (ACM), Amazon Identity and Access Management (IAM), or import a certificate. This certificate will automatically be added to your listener certificate list.

Certificate source

From ACM

From IAM

Import certificate

Certificate (from ACM)

The selected certificate will be applied as the default SSL/TLS server certificate for this load balancer's secure listeners.

boutiquemode.tn

b7cb00ec-e7b9-4d4e-85ef-a22e3f947c49

[Request new ACM certificate](#)

Client certificate handling

Info

Client certificates are used to make authenticated requests to remote servers. [Learn more](#)

Mutual authentication (mTLS)

Mutual TLS (Transport Layer Security) authentication offers two-way peer authentication. It adds a layer of security over TLS and allows your services to verify the client that's making the connection.

Next & Create Load Balancer

Load balancers (1/1) [What's new?](#)

[Refresh](#) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups
☒	alb-boutiquemode	Provisioning	application	Internet-facing	IPv4	vpc-0fd3d704c387ff27d	3 Availability Zones	sg-09fc5679a79

Load balancer: alb-boutiquemode

Details

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

Info

[Refresh](#) [Manage rules](#) [Manage listener](#) [Add listener](#)

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
☐	HTTPS:443	• Forward to target group boutiquemode-tg t2: 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	boutiquemode.tn (Certificate 1...	Off

Step 2: HTTPS Enforcing

4/6

Load balancers (1/1) [What's new?](#)

⌂

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 >

⚙

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security group
☒	alb-boutiquemode	Provisioning	application	Internet-facing	IPv4	vpc-0fd3d704c387ff27d	3 Availability Zones	sg-09fc5679a74

Load balancer: alb-boutiquemode

Details

Listeners and rules

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Attributes

Capacity

Tags

Listeners and rules (1) [Info](#)

⌂

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 >

⚙

☐	Protocol:Port	Default action	Rules	ARN	Security policy	Default SSL/TLS certificate	mTLS
☐	HTTPS:443	• Forward to target group boutiquemode-tg: 1 (100%) Target group stickiness: Off	1 rule	ARN	ELBSecurityPolicy-TLS13-1-2-...	boutiquemode.tn (Certificate I...	Off

EC2 > Load balancers > alb-boutiquemode > Add listener

edited and deleted from the listener.

▶ Load balancer details: alb-boutiquemode

Listener: HTTP:80

A listener checks for connection requests using the protocol and port that you configure. The default action and any additional rules that you create determine how the Application Load Balancer routes requests to its registered targets.

Protocol

Used for connections from clients to the load balancer.

HTTP

Port

The port on which the load balancer is listening for connections.

80

1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☐ Forward to target groups

☒ Redirect to URL

☐ Return fixed response

Redirect to URL [Info](#)

Redirect client requests from one URL to another. You cannot redirect HTTPS to HTTP. To avoid a redirect loop, you must modify at least one of the following components: protocol, port, hostname or path. Components that you do not modify retain their original values.

URI parts

Full URL

Protocol

Used for connections from clients to the load balancer.

HTTPS

Port

The port on which the load balancer is listening for connections.

443

1-65535 or to retain the original port enter #{port}

☐ Custom host, path, query
Select to modify host, path and query. If no changes are made, settings from the request URL are retained.

Status code

301 - Permanently moved

Resulting Ressource Map:

Resource map [Info](#)

View, explore, and troubleshoot your load balancer's architecture.

Overview

Unhealthy target map

Show resource details

alb-boutiquemode

Last fetched seconds ago

⌂

Export

Listeners (2)

HTTPS:443

1 rule

HTTP:80

1 rule

Rules (2)

→ Priority default
Forward to target group

Conditions (If)
If no other rule applies

↪ Priority default
Redirect

Conditions (If)
If no other rule applies

Target groups (1) [Info](#)

Instance, HTTP
boutiquemode-tg

1 target

✔ 1

⊖ 0

⊖ 0

⊖ 0

⊖ 0

Targets (1)

i-0352c615cbb57e906

Port 80

✔ Healthy

5/6

EC2 > Security Groups > sg-0dbe92e65d5ff028 - boutiquemode-web-sg > Edit inbound rules

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

This rule is no longer needed, delete it.

Inbound rules

Info

Security group rule ID

sg-0caf029a088956421

Type

Info

HTTP

▼

Protocol

Info

TCP

▼

Port range

Info

80

▼

Source

Info

Custom

▼

0.0.0.0/0

×

Description - optional

Info

Temporary open HTTP for testing (ALB SG will assure no direct public access to EC2)

Delete

sg-0b74b8fe98d585d24

SSH

TCP

22

Custom

0.0.0.0/0

Temporary EC2 Instance SSH for ot

Delete

Add rule

Cancel

Preview changes

Save rules

EC2 > Security Groups > sg-0dbe92e65d5ff028 - boutiquemode-web-sg > Edit inbound rules

Edit inbound rules

Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Info

Security group rule ID

sg-0b74b8fe98d585d24

Type

Info

SSH

▼

Protocol

Info

TCP

▼

Port range

Info

22

▼

Source

Info

Custom

▼

0.0.0.0/0

×

Description - optional

Info

Temporary EC2 Instance SSH for ot

Delete

-

HTTP

TCP

80

Custom

sg-0b856a02f7779d7e4

Allow HTTP v1 only from ALB

Delete

-

HTTP

TCP

80

Custom

sg-02bcd3894a85b86a8

Allow HTTP v2 only from ALB

Delete

Add rule

Cancel

Preview changes

Save rules